



Next Generation of Policy Frameworks and Modeling

WEEK 2 • JUNE 15-19, 2026

DynareJulia and FPAS Mark II:

Forecasting, Scenario Building, and Risk Analysis

DAY 4 • Thursday, June 18 — Trade-offs & DynareJulia

Interpret policy trade-offs across scenarios and implement them in a modern open-source stack.



Week 2 at a Glance

June 15-19, 2026 — A full policy cycle compressed into five days

DAY 1

Mon · Jun 15

Current Conditions

Data update, weekly briefings, where-is-the-economy-now workflow.

DAY 2

Tue · Jun 16

Market Reference & Case A/B

Scenario construction: market reference, Case A and Case B narratives.

DAY 3

Wed · Jun 17

Taxonomy & Risks

Full taxonomy of scenarios, dark corners, risk assessment.

DAY 4

Thu · Jun 18

Trade-offs & DynareJulia

Policy trade-off analysis and DynareJulia implementation.

DAY 5

Fri · Jun 19

MPR & Communication

Drafting the MPR, Python pipelines, transparency and Board communication.



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What is a policy trade-off?

Every scenario implies a sequence of compromises between competing objectives

THE COMPETING OBJECTIVES

Inflation vs activity

- Bringing inflation back to target faster = more output loss today
- Smoothing the output path = more inflation persistence
- Anchoring expectations = willingness to accept short-term costs
- Financial stability and credibility weigh implicitly on the choice

HOW FPAS MARK II FRAMES IT

Trade-off across scenarios

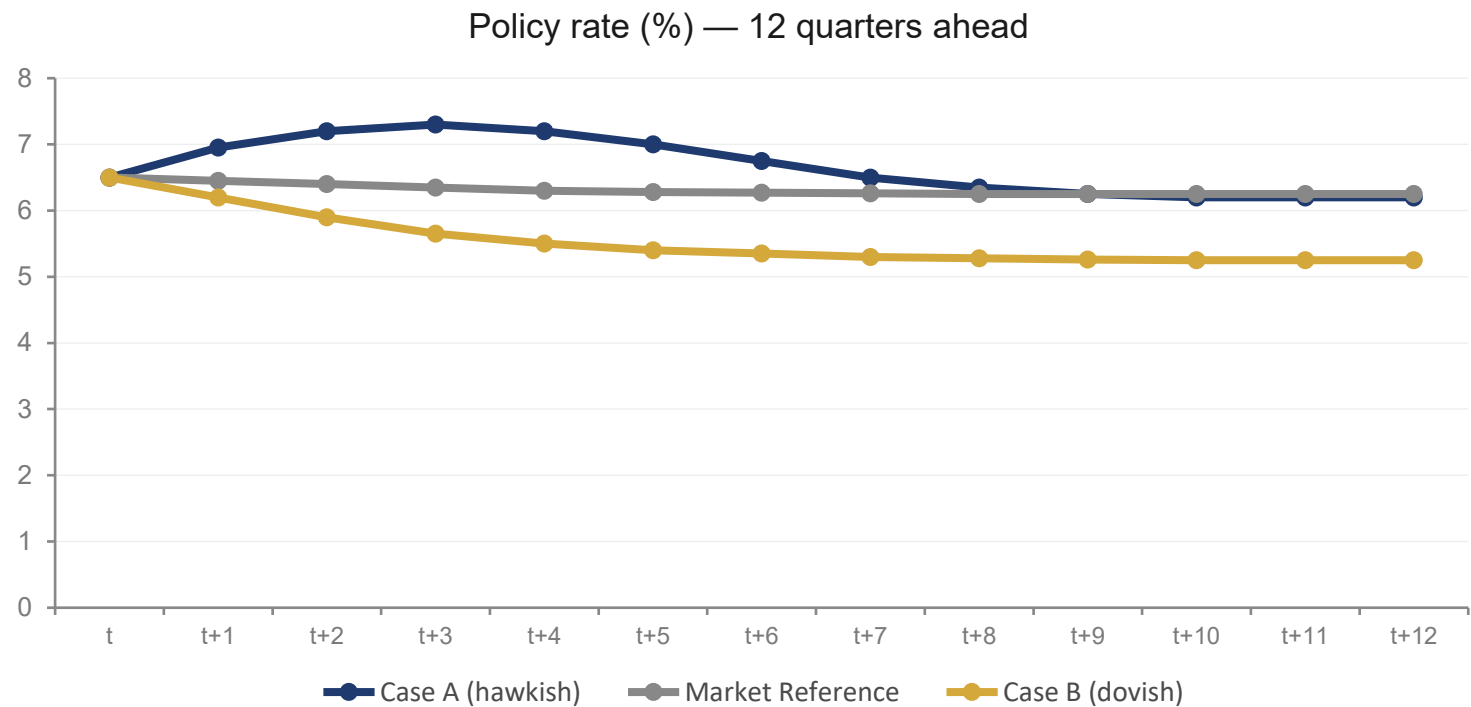
- Compare inflation, output, and rate paths across Case A and Case B
- Evaluate the losses associated with each scenario's policy response
- Design a policy that performs acceptably across all relevant scenarios
- Communicate the trade-off explicitly — not hide it behind a baseline



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Policy rate paths: Armenia Q1 2026

Three paths — same starting point, different destinations



READING THE CHART

Case A

Rate peaks around +70-80bp above market, then converges to neutral.

Market Reference

Gradual descent to ~6.25% — the market-priced neutral level.

Case B

Faster and deeper easing — below 5.5% by horizon end.



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Enter DynareJulia

A modern open-source platform for implementing FPAS Mark II scenarios

WHAT IT IS

DynareJulia

- A rewrite of Dynare in the Julia programming language
- Built on decades of Michel Juillard's work on Dynare
- Current version: v0.9.18 — actively developed
- Solves, simulates, and analyzes DSGE, OLG, and other structural models

WHY IT MATTERS

Transparent & reproducible

- Open-source: no licensing barriers for public policy institutions
- Julia's speed: significantly faster than MATLAB for nonlinear solves
- Reproducible workflows: share the model, not just the results
- Community-backed: public docs, User Guide, active repository



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What you can do with DynareJulia

Four families of models, one consistent syntax and solver

Simple Linear Models

Semi-structural QPMs for baseline scenario work and calibration exercises.

Output gap, Phillips curve, Taylor rule — the workhorse for inflation targeting.

Endogenous Credibility (ENDOCRED)

Nonlinear models where credibility itself is a state variable.

Captures FPAS Mark II features: credibility erodes when inflation drifts, harder to rebuild.

DSGE with Occasionally Binding Constraints

Handles the effective lower bound, collateral constraints, and other nonlinearities.

Essential for Case B scenarios near dark corners.

GIMF-Style Global Models

Multi-country framework for monetary-fiscal interactions and international spillovers.

Useful for Day 4's structural policy analysis.



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DynareJulia workflow for a scenario

From .mod file to publication-ready output

01

Define

Write the model in a .mod file: variables, parameters, equations, shocks.

02

Solve

DynareJulia computes the steady state and solves the linear or nonlinear policy functions.

03

Simulate

Impose the scenario's shock paths and generate the implied trajectory for all variables.

04

Compare

Stack Case A, Case B, and Market Reference simulations. Export for charts and tables.

User Guide: asyakostanyan.github.io/DynareJulia-User-Guide/ · Git: github.com/DynareJulia/Dynare.jl



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End-of-day deliverable

By the end of Day 4, each team produces a set of quantified scenarios ready for the MPR

1

Three quantified paths

Case A, Case B, and Market Reference — all simulated through DynareJulia with consistent shock calibration.

2

Policy trade-off analysis

Explicit comparison of inflation, output gap, and rate paths. Losses evaluated across scenarios.

3

Chart pack ready

Exported simulation output formatted for the MPR chart pack: rate path, inflation fan, output gap, exchange rate.

4

Model package

Clean .mod file, calibration notes, and reproducible scripts — shareable and extendable for the next round.



Great Things are Coming.



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